

GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER V

Name of Department: - Computer Science and Engineering

Course Code:		TCS509		Course Title:		Machine Learning	
Credits:	3	Semester:		V	Category of Course:		DSC
TEACHING PERIODS (In hours per week)			TW+SL (in hours per semester)		Relative Weight		
L	T	P	45		CIE	MSE	ESE
3	0	0			25	25	50
Examination Duration (Hrs):			Theory			Practical	
			3			0	

Pre-requisite: TMA 101, TMA 201

Course Outcome:	After completion of the course the students will be able to: CO1 Understand and explain the fundamental concepts of machine learning, including learning problems, inductive bias, learning system design, and model validation techniques. CO2 Apply and evaluate supervised learning algorithms, including regression, classification, support vector machines, and nearest neighbour methods for predictive modelling. CO3 Design and evaluate decision tree-based, ensemble, and probabilistic learning models using appropriate performance measures for classification tasks. CO4 Apply clustering, dimensionality reduction, and feature selection techniques to analyze and improve machine learning models. CO5 Design and implement neural network models and understand the fundamentals of deep learning and reinforcement learning using modern machine learning frameworks.
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Details of the Course:

UNIT	CONTENTS	Contact Hrs
Unit - I	Learning Problems, Designing a Learning System, Perspectives and Issues in Machine Learning Concept Learning and the General-to-Specific Learning: Concept Learning Task, Concept learning as Search, version spaces, Inductive Bias Validation Techniques: Hold out, K-Fold Cross Validation, Leave one out, Bootstrapping.	7
Unit - II	Curve fitting, Linear Regression, Logistic Regression, Regularization, Mean Absolute Error, Mean Squared Error, Root Mean Square Error, Bias and Variance, Overfitting and Underfitting Support Vector Machine, K-Nearest Neighbour, Distance Weighted Nearest Neighbour, Locally Weighted Regression	10
Unit – III	Decision Trees, Hypothesis Space Search in Decision Tree Learning, Inductive Bias, Random Forest, Ensemble Learning: Bagging, Boosting and Stacking Bayes Theorem, Naive Bayes, Maximum Likelihood Estimation (MLE), Maximum A posteriori Estimation (MAP), Bays Optimal Classifier, Bayes Inference.	10
Unit – IV	K-means, Silhouette Scores, Elbow Method, Hierarchical: Agglomerative & Divisive Clustering, Fuzzy Clustering, DBSCAN, Error Analysis in Clustering Curse of Dimensionality, Dimensionality reduction techniques, Principal Component Analysis, Linear Discriminant Analysis, Feature Selection Methods: Forward and Backward	9
Unit – V	Neural Network, Perceptron Algorithm, Multilayer Networks, Backpropagation Algorithm, Gradient Decent Algorithm, Learning Rate, Introduction to Reinforcement Learning, Introduction to Deep Learning: Convolution Neural Network, Autoencoders, Resnet Deep Learning Tools: Pytorch and TensorFlow	9
Total		45

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
Tom M. Mitchell,	Machine Learning –	1st Edition	Mc Graw Hill Publisher	2017
R, Duda, P. Hart and D. Stork,	Pattern classification –	2nd.Edition	Wiley Publisher	2007
– J. Friedman, R. Tibshirani and T. Hastie,	The Elements of Statistical Learning	3rd.Edition	Springer Publisher	2017
A. Courville, I. Goodfellow, Y. Bengio,	Deep Learning –	2nd.Edition	MIT Press	2016

Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
C. Bishop,	Pattern Recognition and Machine Learning –	2nd.Edition	Springer Publisher	2016

E. Alpaydin,	Introduction to Machine Learning –	3rd. Edition	PHI Publisher	2015
T M. Cover, J A. Thomas,	Elements of Information Theory	2nd.Edition	Wiley Publisher	2006